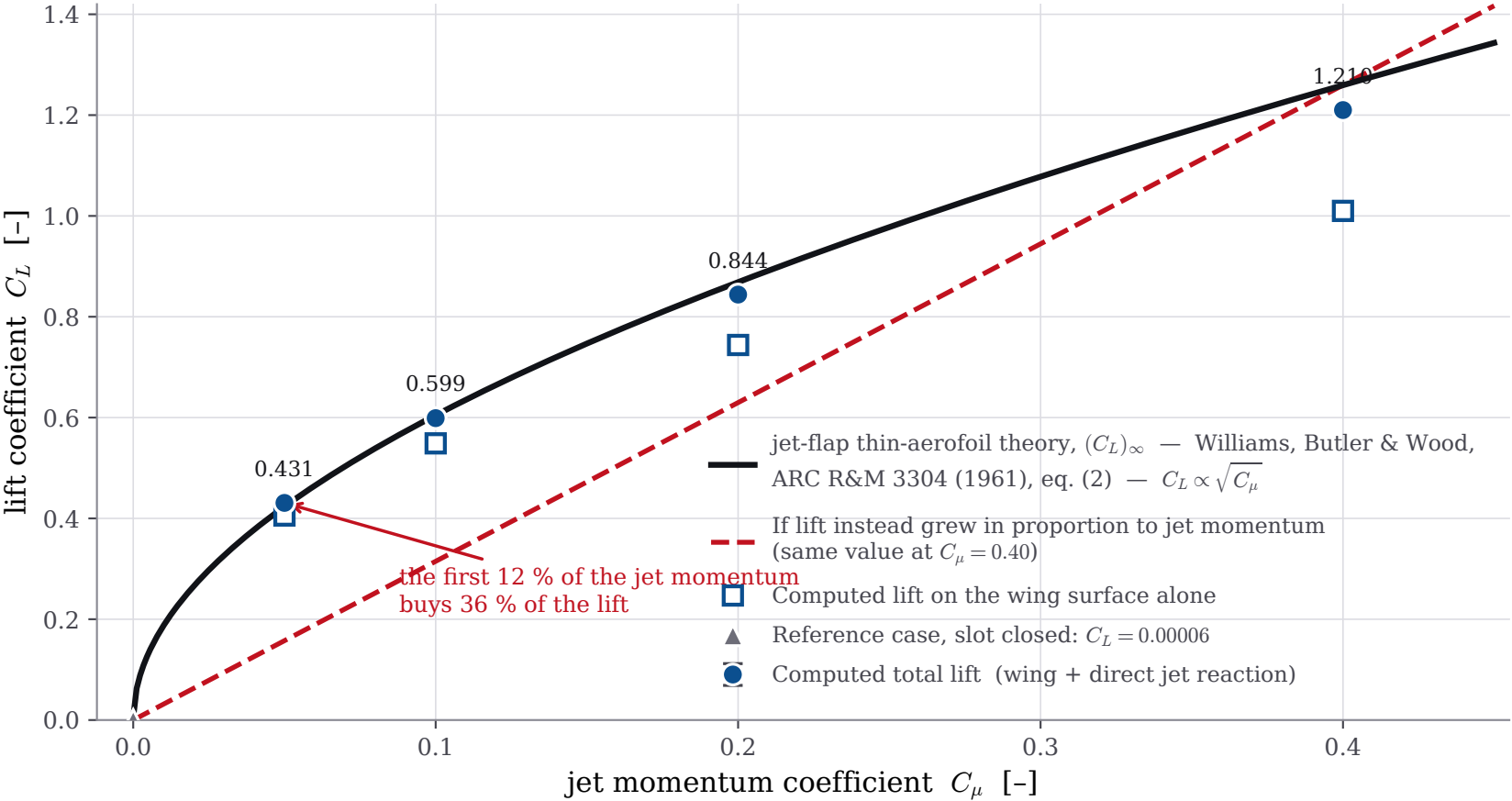


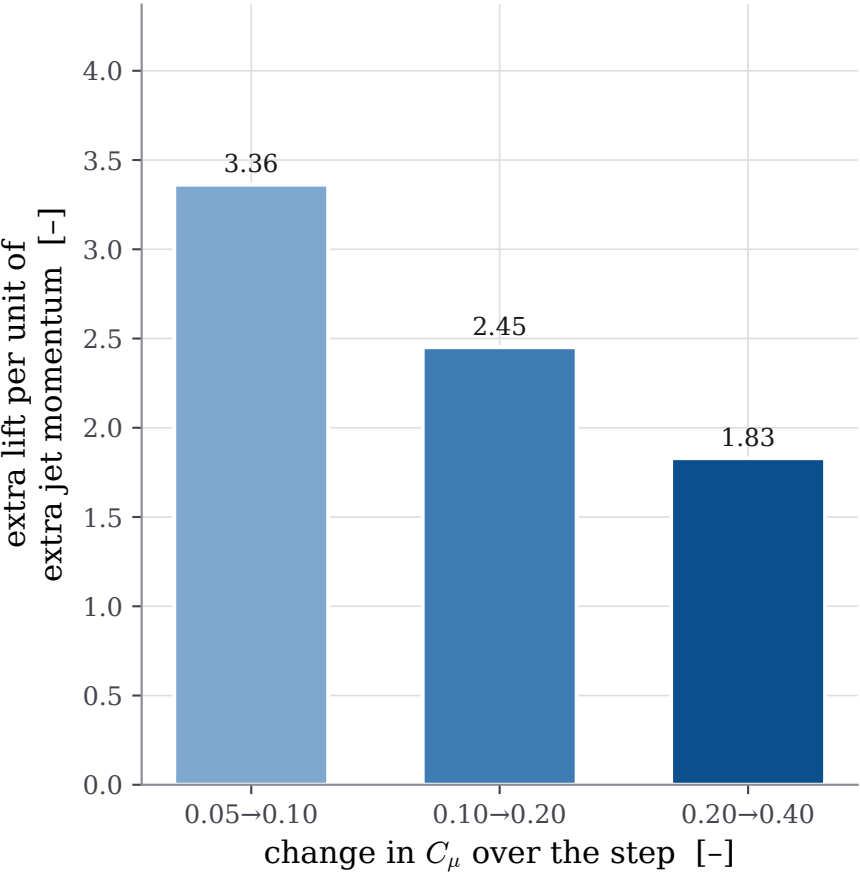
PRELIMINARY — EXPLORATORY CALCULATION, NOT A VALIDATED RESULT

Two-dimensional wing section, 1.0 m chord, symmetric, at zero incidence in a 10 m s⁻¹ stream. A 5.0 mm slot at the trailing edge blows a sheet of air aimed 30° below the chord line. C_μ is the jet's momentum divided by the oncoming stream's dynamic pressure times the chord. All five calculations share one mesh of 39 984 cells. The four blown cases differ from one another only in how hard the slot blows, so the differences among those four are caused by the blowing and by nothing else. The fifth is the reference case, with the slot closed — not the same wing with the jet turned down to zero.

Lift grows as the SQUARE ROOT of jet momentum, not in proportion to it



Each extra unit of blowing buys less than the one before



WHAT YOU SHOULD KNOW ABOUT THESE NUMBERS

- None of these calculations reached the convergence target that was fixed before they were run (all five solution channels below 1×10^{-6}). The turbulence-energy channel is the slowest everywhere, and it gets worse as blowing increases — from 3.5×10^{-6} with no jet to 1.5×10^{-4} at the strongest jet, a factor of 43. Strong blowing makes this flow genuinely hard to converge; that is a property of the physics, not of any one calculation.
- Over the final 4,000 iterations the lift still moves by up to 3.4×10^{-5} at the worst row. That is a settling indicator, not convergence. Those bars are drawn but are smaller than the symbols.
- Mesh sensitivity has NOT been quantified for these points. No refinement study was run, so no grid uncertainty is claimed.
- Treat every number here as indicative. It is not a validated result and it has not been compared against experiment.